

Capstone Project – Step 5 BindFlow API Specification

1. General Guidelines

- All protected routes require a valid JWT token in the header:
- Authorization: Bearer <my_token>
- Role-based access will be enforced using middleware.
- Pagination will be supported on list endpoints using page and limit query parameters.
- Google Books API will be called through the backend (proxy).

2. CORS Implementation

CORS will be implemented on the backend so the React frontend can communicate with the Express API.

Installation: npm install cors

In JavaScript CORS will be required via → `const cors = require('cors');`

An example of a portion of the code to be used can be observed below:

```
const cors = require('cors');

const allowedOrigins = process.env.CLIENT_URL
  ? process.env.CLIENT_URL.split(',')
  : ['http://localhost:5173'];

app.use(cors({
  origin: allowedOrigins,
  credentials: true,
  methods: ['GET', 'POST', 'PUT', 'PATCH', 'DELETE'],
  allowedHeaders: ['Content-Type', 'Authorization']
}));
```

3. Authentication EndPoints

POST/api/auth/register

- Description: Register a new user (default role = client)
- Auth Required: No
- Request Body: name, email, password, phone (optional), businessName (optional)
- Success Response: User object + JWT token

POST /api/auth/login

- Description: Login user
- Auth Required: No
- Request Body: email, password
- Success Response: User object + JWT token

GET /api/auth/me

- Description: Get currently logged-in user
- Auth Required: Yes
- Success Response: Current user object

POST /api/auth/logout

- Description: Logout user
- Auth Required: Yes
- Success Response: Success message

4. Book Search Endpoints

- Description: Search for books using Google Books API (proxied through backend)
- Auth Required: Yes
- Query Parameter: q (search term)
- Success Response: Array of books (title, author, isbn, coverImage, googleBooksId)

5. Order Endpoints

Client Routes

POST /api/orders

- Description: Client submits a new bookbinding order
- Auth Required: Yes (Client role)
- Request Body: book object + specifications object
- Success Response: Newly created order

GET /api/orders/my

- Description: Get all orders belonging to the logged-in client
- Auth Required: Yes (Client role)
- Query Parameters: page, limit, status (optional)
- Success Response: Paginated list of client's orders

GET /api/orders/:id

- Description: Get details of one specific order (client can only see their own)
- Auth Required: Yes (Client role)
- Success Response: Full order details

Staff / Admin Routes GET /api/orders

- Description: Get all orders in the system (with filters and pagination)
- Auth Required: Yes (Staff or Admin)
- Query Parameters: page, limit, status, search, clientId (optional)
- Success Response: Paginated list of orders

GET /api/orders/:id

- Description: Get full details of any order
- Auth Required: Yes (Staff or Admin)
- Success Response: Complete order object

PATCH /api/orders/:id/quote

- Description: Create or update the quote for an order
- Auth Required: Yes (Staff or Admin)
- Request Body: totalPrice, breakdown (optional)
- Success Response: Updated order with quote

PATCH /api/orders/:id/status

- Description: Update the production status of an order
- Auth Required: Yes (Staff or Admin)
- Request Body: status (e.g. quoted, in-production, ready, delivered)
- Success Response: Updated order

POST /api/orders/:id/deliveries

- Description: Record a partial delivery for an order
- Auth Required: Yes (Staff or Admin)
- Request Body: deliveryDate, quantityDelivered, notes (optional)
- Success Response: Updated order with new delivery

PATCH /api/orders/:id/payment

- Description: Mark payment as received or verified
- Auth Required: Yes (Staff or Admin)
- Request Body: status, amount (optional), method (optional), notes (optional)
- Success Response: Updated order with payment info

6. Stretch / Future Endpoints (Evaluating still if this is going to be implemented)

POST /api/orders/:id/pdf

- Description: Generate PDF quote or invoice
- Note: Can also be handled on the frontend using jsPDF

GET /api/dashboard/stats

- Description: Return dashboard statistics and counts
- Note: For analytics stretch goal

7. Final Notes

- Protected routes and role-based middleware will be implemented.
- All data validation will be handled using Mongoose schemas.
- CORS is configured using the cors package with support for both development and production environments.
- The API structure can be adjusted during development as needed.